

Student: Variant B (PDF Submission)

The cellular boundary consists of a fluid lipid structure that exhibits strict permeability. Its main job is to maintain homeostasis by modulating what enters and exits the cytoplasm. Tiny, non-polar substances cross the boundary spontaneously without energy. Bulkier compounds and electrolytes, however, depend on specialized transmembrane channels for transit, often expending energy to move against concentration gradients. Additionally, surface glycoproteins allow for cellular communication and hormone recognition, and lipid sterols preserve the flexibility of the wall.